

Can compensation policy increase tolerance toward wild boars?

Understanding the compensation policy impact on local resident's tolerance towards wild boars in Giant Panda National Park, China

Human–wildlife conflict (HWC) is widely recognized as one of the most pressing challenges in contemporary wildlife conservation (König et al., 2020). It refers to interactions between humans and wildlife that result in negative outcomes (Madden, 2004), including crop damage, livestock depredation, and threats to human safety (Zimmermann et al., 2020). Beyond its ecological implications, HWC is fundamentally a social issue, as its persistence and escalation are closely linked to human attitudes toward wildlife.

Within HWC mitigation research, tolerance has emerged as a key indicator of coexistence potential (Expósito-Granados et al., 2019). Tolerance is commonly defined as the ability and willingness of individuals to absorb actual or perceived costs associated with living alongside wildlife (Kansky & Kidd, 2024). Previous studies show that tolerance is shaped not only by tangible costs and benefits, such as economic losses or compensation, but also by intangible dimensions, including emotional responses, perceived risks, and social meanings attached to wildlife (Kansky & Knight, 2014).

Compensation policies are widely used to mitigate human–wildlife conflict (HWC) by offsetting economic losses and improving local tolerance toward wildlife (Hemminger et al., 2025; Kichloo et al., 2023). In China, wild boar (*Sus scrofa*) populations have increased rapidly in recent decades, leading to intensified crop damage and frequent compensation claims, especially in and around national parks. Despite the widespread implementation of compensation schemes, preliminary empirical evidence from my research suggests that financial compensation does not significantly increase tolerance toward wild boar. This finding contradicts common policy assumptions and raises important questions about the social effectiveness of compensation-based governance instruments. Understanding why compensation fails to translate into tolerance is crucial for improving coexistence-oriented wildlife management and for designing socially sustainable conservation policies.

Research Question

This project addresses the following research questions:

1. Does compensation for wild boar damage have a direct effect on tolerance toward wild boar, or is its influence mediated by other social and psychological factors?
2. What roles do tangible costs, intangible costs, and intangible benefits play in shaping tolerance, and how do these factors interact with compensation experience?
3. How robust are these relationships under different model specifications and across groups with and without compensation experience?

Methodological Challenges

Key methodological questions include:

1. Model performance and specification
 - a. How can the structural model be improved (e.g. path structure, mediation, moderation)?
 - b. Should certain constructs be merged, re-specified, or modeled formatively vs. reflectively?
2. Measurement model refinement
 - a. Assessment and improvement of construct validity and reliability
 - b. Evaluation of indicator selection and potential redundancy
3. Robustness and alternative modeling strategies
 - a. Sensitivity checks and alternative model comparisons
 - b. Discussion of whether complementary approaches (e.g. multi-group analysis or simplified models) are appropriate

Project partners

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